Study of One-Dimensional Satellite Attitude Control Using Reaction Wheels

Srujana Patil

June 15, 2026

Contents

Chapter 1

Fundamentals of Satellite Attitude Control

1.1 Introduction to Satellite Attitude Control

Satellite attitude control refers to controlling the orientation of a spacecraft in space. Proper attitude control is necessary for applications such as communication, Earth imaging, navigation, and scientific observations.

One of the most commonly used methods for satellite attitude control is the reaction wheel system. Reaction wheels allow the spacecraft to rotate without using propellant, making them suitable for long-duration missions.

The objective of this project is to study the dynamics and control of a one-dimensional satellite attitude control system using a reaction wheel.

1.2 Reaction Wheel

Reaction wheels are commonly used in small spacecraft to provide precise three-axis attitude control. *They operate by storing angular momentum along the wheel spin axis, causing the spacecraft body to rotate in the opposite direction due to conservation of angular momentum.* During spacecraft deployment, small unwanted angular rates known as tipoff momentum are often induced, and reaction wheels help reduce these rates by absorbing and storing the momentum within the spinning wheels.

The selection of reaction wheels depends on several factors such as the spacecraft moment of inertia, required momentum storage capacity, and desired slew rate. However, each wheel has a maximum rotational speed, and once this limit is reached, the wheel becomes saturated and can no longer provide additional torque or momentum storage along that axis. To prevent saturation, the stored momentum must be periodically removed through a process known as desaturation or momentum dumping using external torque-producing actuators such as thrusters or magnetic torquers.

A three-axis stabilized spacecraft typically requires three reaction wheels mounted orthogonally, while four-wheel skewed or pyramid configurations are commonly used to provide redundancy in case of wheel failure. In skewed configurations, the wheels are mounted at angles such that multiple wheels contribute to control across different axes, allowing reduced but continued control capability even if one wheel fails.

Although reaction wheels are highly effective for precision pointing, they can introduce disturbances due to static and dynamic imbalances, producing unwanted torques, vibrations, and jitter. To minimize these effects, wheels are generally placed close to the spacecraft center of gravity. Reaction wheels may also generate magnetic interference because of the electromagnetic components and ferrous materials used in their construction, requiring careful placement away from sensitive instruments such as magnetometers.

Another important consideration is wheel operation near zero speed, where performance may degrade during transitions between positive and negative rotation. To reduce jitter and improve control precision, some missions maintain the wheels at a constant biased speed, although this reduces the total available momentum storage capacity.

1.3 Conservation of Angular Momentum

The motion of the spacecraft-reaction wheel system is based on the principle of conservation of angular momentum.

For a rotating rigid body,

$$H = I\omega$$

where:

- H = angular momentum
- I = moment of inertia
- ω = angular velocity

If no external torque acts on the system,

$$\frac{dH}{dt} = 0$$

For the spacecraft and reaction wheel together,

$$I_s\dot{\omega}_s + I_w\dot{\omega}_w = 0$$

where:

- I_s = spacecraft moment of inertia
- I_w = reaction wheel moment of inertia
- ω_s = spacecraft angular velocity
- ω_w = reaction wheel angular velocity

This shows that when the reaction wheel speeds up in one direction, the spacecraft rotates in the opposite direction in order to conserve angular momentum.

1.4 Equation of Motion

Including damping effects, the equation of motion of the spacecraft can be written as:

$$I_s\ddot{\theta} + b\dot{\theta} = \tau$$

where:

- I_s = spacecraft moment of inertia
- b = damping coefficient
- τ = control torque produced by the reaction wheel
- θ = angular displacement

1.5 Transfer Function and Stability Analysis

Taking the Laplace transform assuming zero initial conditions,

$$I_s s^2\Theta(s) + bs\Theta(s) = T(s)$$

Rearranging,

$$\frac{\Theta(s)}{T(s)} = \frac{1}{I_s s^2 + bs}$$

This transfer function represents the rotational dynamics of the spacecraft.
the poles are obtained from-

$$s(I_s s + b) = 0$$

Thus

$$s_1 = 0$$

$$s_2 = -\frac{b}{I_s}$$

For $b > 0$, one pole lies at the origin and the other lies in the left half plane. Therefore, the uncontrolled system is marginally stable.

1.6 Spacecraft Parameters and Moment of Inertia

For demonstration purposes, a microsatellite configuration is considered.
For a cuboidal spacecraft, the moment of inertia is given by:

$$I = \frac{1}{12}m(a^2 + b^2)$$

where:

- m = spacecraft mass
- a, b = spacecraft dimensions

This relation shows that the moment of inertia changes with both spacecraft mass and dimensions.

1.7 Factor of Safety

In spacecraft hardware design, a factor of safety is used to account for uncertainties such as launch vibrations, material imperfections, unexpected loading conditions, and manufacturing tolerances.

In this project, a factor of safety of 3 is considered for hardware sizing calculations.

The spacecraft mass considered for practical analysis is approximately $70\,kg$, and conservative design calculations are performed using the specified factor of safety.

1.8 Damping Coefficient

The damping coefficient b represents energy losses in the system due to effects such as friction, structural vibrations, and internal mechanical losses.

In practical spacecraft systems, the damping coefficient may vary with time because of changing operating conditions, temperature variations, and component wear. Therefore, b may be considered time varying in realistic models.

1.8.1 Time varient damping

In a practical satellite attitude control system, the damping coefficient is not strictly constant and varies with time due to mechanical effects such as bearing friction variation, lubrication degradation, thermal expansion, and component aging. Therefore, the classical assumption of a Linear Time-Invariant (LTI) system becomes invalid, and the system behaves as a Linear Time-Varying (LTV) system. To model this behavior, the damping coefficient was represented as an exponentially varying function of time:

$$b(t) = b_0 e^{-\alpha t}$$

where b_0 is the initial damping coefficient and α is the damping degradation constant. The satellite rotational dynamics therefore become:

$$I\ddot{\theta}(t) + b(t)\dot{\theta}(t) = \tau(t)$$

Since variations in damping directly affect transient response characteristics such as overshoot and settling time, a fixed derivative gain was insufficient for maintaining consistent closed-loop performance. To compensate for the changing physical damping, an adaptive derivative gain was introduced:

$$K_d(t) = K_{d0}\frac{b_0}{b(t)} = K_{d0}e^{\alpha t}$$

where K_{d0} is the nominal derivative gain. This adaptive strategy increases derivative action when physical damping decreases, thereby preserving effective system damping and maintaining stable attitude tracking performance. Numerical simulation using time-domain state-space integration demonstrated that the adaptive PID controller successfully maintained a nearly unchanged step response despite the presence of time-varying damping.

1.9 Limitations of Reaction Wheels

- **Limited Torque Output**
- **Momentum Saturation**
- **Mechanical Wear**
- **Power Consumption**
- **Vibration and Imbalance**

Chapter 2

Sensors and State Estimation

2.1 Inertial Measurement Unit (IMU)

An Inertial Measurement Unit (IMU) is used to measure the motion and orientation of the spacecraft.

An IMU mainly contains:

- Gyroscopes
- Accelerometers

These sensors provide important feedback for closed-loop attitude control.

2.2 Gyroscope

A gyroscope is a sensor used to measure the angular velocity of the spacecraft about its axes.

$$\omega_x, \omega_y, \omega_z$$

In this project, the gyroscope is used to detect how fast the satellite is rotating. The measured angular velocity can then be integrated to estimate the angular position θ of the satellite.

2.3 Accelerometer

An accelerometer is a sensor that measures acceleration acting along its axes.

For a three-axis accelerometer, acceleration is measured along the x, y, and z directions.

When the sensor is stationary and only gravity acts on it:

$$a_x = 0$$

$$a_y = 0$$

$$a_z = -9.8\, m/s^2$$

assuming the z-axis is aligned vertically.

If the sensor is tilted, the effect of gravity gets distributed among the different axes. As a result, the values of a_x, a_y, and a_z change.

This property allows the accelerometer to estimate the orientation of the spacecraft relative to the gravity direction.

In this project, accelerometer measurements are combined with gyroscope measurements to improve the estimation of the satellite angle θ. Although gyroscopes provide accurate short-term rotational information, small errors accumulate over time. Accelerometers help correct this drift and improve the estimation of the satellite orientation.

2.4 Kalman Filter

In practical systems, sensor measurements contain noise, drift, and small errors. A Kalman filter is a mathematical estimation algorithm used to combine data from multiple sensors and obtain a more accurate estimate of the system state.

The *Kalman filter continuously predicts the satellite state and corrects the prediction using sensor measurements through recursive mathematical calculations based on linear algebra and probability.* This produces a more stable and accurate estimate of the satellite orientation.

In this project, the Kalman filter combines gyroscope and accelerometer measurements to estimate the satellite orientation angle θ. The gyroscope provides accurate short-term rotational information but suffers from drift over time, while the accelerometer provides a long-term reference but is sensitive to noise and disturbances.

The estimated angle is compared with the reference angle θ_{ref}, and the resulting error signal is used by the controller to generate the required control torque for the reaction wheel system.

The estimated angle is then compared with the reference angle θ_{ref}. The difference between the two gives the error signal:

$$e(t) = \theta_{ref}(t) - \theta(t)$$

This error signal is fed into the controller, which generates the required control torque for the reaction wheel so that the satellite rotates toward the desired reference orientation.

Chapter 3

Diagrams and Simulations

3.1 Simulation Parameters

For this project, the satellite is assumed to be a simplified cubical microsatellite of dimensions $100 \times 100 \times 100$ cm with a total mass of $m = 70, kg.$

Using the standard moment of inertia relation for a cuboidal body,

$I = \frac{1}{12} m a^2$

the spacecraft moment of inertia is approximated as $I \approx 12, kg \cdot m^2.$

For all MATLAB simulations, the damping coefficient is taken as $b = 2.$

These parameters are used throughout the simulations to study the closed-loop attitude response and controller performance of the reaction wheel system.

3.2 Simplest Closed loop with tuned PID

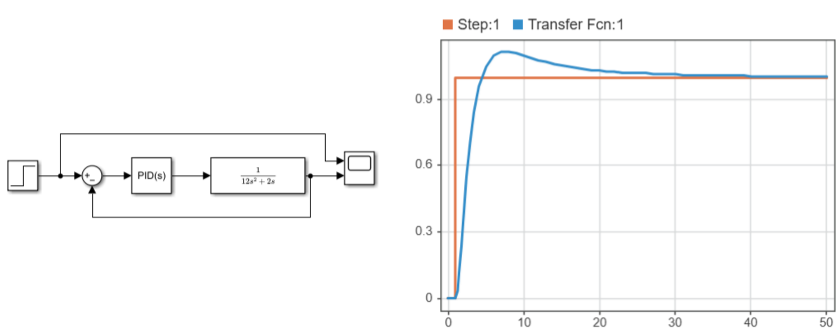

Figure 3.1: simplest closed loop system with step input and auto tuned PID using Matlab simulink

Using MATLAB PID tuning tools, the initial tuned controller parameters obtained for the closed-loop system were:

$K_p = 1.81891$
$K_i = 0.093815$
$K_d = 6.11144$

3.3 Controller Design and Performance Criteria

However, for further analysis and simplicity, a PD controller is considered in this project. Since the spacecraft plant already contains the integrator term $\frac{1}{s}$, acceptable steady-state performance can still be achieved without explicitly using integral control.

The spacecraft plant transfer function is given by:

$$G(s) = \frac{1}{12s^2+2s}$$

The PD controller transfer function is:

$$C(s) = K_p + K_d s$$

The open-loop transfer function therefore becomes:

$$G_{OL}(s) = \frac{K_p+K_d s}{12s^2+2s}$$

The closed-loop transfer function of the system is therefore obtained as

$$T(s) = \frac{K_d s + K_p}{12s^2 + (2 + K_d)s + K_p}$$

For a unit step input, the output response is given by

$$Y(s) = \frac{1}{s} \frac{K_d s + K_p}{12s^2 + (2 + K_d)s + K_p}$$

Applying partial fraction decomposition and taking the inverse Laplace transform yields

$$y(t) = 1 - e^{-at} \cos(\omega_d t) + \frac{K_d - 2}{24\omega_d} e^{-at} \sin(\omega_d t)$$

where

$$a = \frac{2 + K_d}{24}$$

and

$$\omega_d = \sqrt{\frac{K_p}{12} - a^2}$$

Defining

$$M = \frac{K_d - 2}{24\omega_d}$$

the response can be rewritten as

$$y(t) = 1 - \sqrt{1 + M^2}\, e^{-at} \cos\left(\omega_d t + \tan^{-1}(M)\right)$$

The peak time is obtained by setting

$$\frac{dy(t)}{dt} = 0$$

which gives

$$T_p = \frac{\pi - \tan^{-1}(M) + \tan^{-1}\left(\frac{a}{\omega_d}\right)}{\omega_d}$$

Substituting T_p into the response equation, the overshoot is obtained as

$$M_p = \sqrt{1 + M^2}\, e^{-aT_p} \frac{\omega_d}{\sqrt{\omega_d^2 - a^2}}$$

Similarly, the rise time and settling time are given by

$$T_r = \frac{\tan^{-1}(1/M)}{\omega_d}$$

and

$$T_s = \frac{96}{2 + K_d}$$

respectively.

The controller gains K_p and K_d were determined using the Levenberg–Marquardt (LM) optimization method. Initially, a target overshoot of 5 percent and a transient response constraint of

$$T_s - T_r = 2\ s$$

were imposed. However, repeated LM optimization and brute-force parameter searches did not yield a feasible solution. The minimum achievable overshoot under this constraint was found to be approximately 15.8 percent.

To investigate whether a relaxation of the transient response requirement could improve the overshoot performance, the constraint was modified to

$$2 \leq T_s - T_r \leq 4\ s$$

A second optimization was then carried out. The minimum overshoot obtained within this range was approximately 14.6 percent. Since the desired overshoot of 5 percent could not be achieved even after relaxing the transient response constraint, it was concluded that the specified overshoot requirement is not attainable for the chosen plant-controller configuration.

3.4 Step Maneuver Strategy for Large Angle Rotations

The controller gains obtained using the Levenberg–Marquardt (LM) optimization method were

$$K_p = 75.699 \qquad K_d = 36.680$$

These gains were found to provide satisfactory transient response characteristics while maintaining stable spacecraft attitude control.

To evaluate the controller performance over the complete operating range, a Python script was developed to generate a table of overshoot values for reference angles ranging from 0° to 180°. Using the optimized controller gains, the overshoot in degrees was calculated for each commanded attitude angle. The generated results showed that the overshoot increased approximately linearly with the commanded reference angle.

Analysis of the generated data indicated that the maximum reference angle which satisfied the design requirement of an overshoot less than 5° was approximately

$$\theta_{ref,max} = 31.73^\circ$$

To provide a safety margin and simplify implementation, the maximum allowable single maneuver angle was selected as

$$\theta_{step} = 30^\circ$$

For a 30° maneuver, the predicted overshoot was approximately 4.73°, which satisfies the specified overshoot requirement.

Consequently, attitude commands greater than 30° are not executed as a single maneuver. Instead, the desired rotation is divided into a sequence of smaller commands, each not exceeding 30°. This approach ensures that every individual maneuver remains within the allowable overshoot limit.

For example, an 80° attitude change is executed as

$$30^\circ \rightarrow 30^\circ \rightarrow 20^\circ$$

Similarly,

$$180^\circ = 30^\circ + 30^\circ + 30^\circ + 30^\circ + 30^\circ + 30^\circ$$

The same controller gains are retained throughout all maneuver steps:

$$K_p = 75.699 \qquad K_d = 36.680$$

Since the spacecraft model is linear, the controller gains are independent of the commanded attitude angle. Therefore, gain scheduling is not required and the same controller can be used throughout the entire operating range.

The primary advantage of the proposed strategy is that it guarantees compliance with the overshoot requirement while avoiding additional controller tuning. Furthermore, implementation is straightforward since only the reference command is updated between successive maneuvers.

The main disadvantage is an increase in the total maneuver time, as large-angle rotations must be completed through multiple smaller commands. However, this trade-off was considered acceptable because it ensures improved pointing accuracy and prevents excessive overshoot during large-angle attitude maneuvers.

3.4.1 Implementation of Step Maneuver Strategy in Simulink

The analytical study showed that the optimized controller gains

$$K_p = 75.699, \qquad K_d = 36.680$$

produced a minimum overshoot of approximately 15.76% while satisfying the transient response constraint of $T_s - T_r \approx 2$ s. Since the desired overshoot specification of 5% could not be achieved through gain tuning alone, a step maneuver strategy was adopted. Using the analytical overshoot value, it was found that a single maneuver could satisfy the 5° overshoot requirement only up to approximately 31.73°. Therefore, a maximum step size of 30° was selected for implementation.

The effectiveness of the proposed approach was verified in Simulink. For a direct 80° attitude command, the spacecraft reached a peak angle of approximately 97°, corresponding to an overshoot of 17°, with a settling time of approximately 5 s. When the same maneuver was executed using the proposed step maneuver strategy, the command was divided into three smaller rotations of 30°, 30°, and 20°. The corresponding peak responses were approximately 36.4°, 66.4°, and 84°, resulting in significantly lower overshoot for each individual maneuver. Although the overall settling time increased to approximately 9 s, the reduction in overshoot produced a considerably smoother and more accurate attitude response. Hence, the step maneuver strategy was selected for the final implementation.

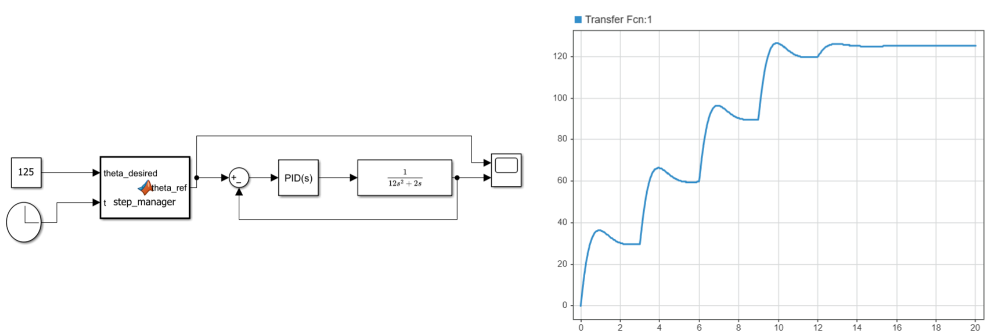

Figure 3.2: simplest closed loop system with step input and auto tuned PID using Matlab simulink

Figure 3.2 shows the Simulink model used to implement the proposed step maneuver strategy. A Constant block provides the desired spacecraft attitude angle $\theta_{desired}$, while a Clock block supplies the simulation time to a MATLAB Function block named `step_manager`.

The `step_manager` block automatically divides any commanded rotation into increments of 30°. For example, a commanded rotation of 125° is internally converted into the sequence

$$30° \rightarrow 60° \rightarrow 90° \rightarrow 120° \rightarrow 125°$$

with each reference update occurring every 3 s. This eliminates the need to manually modify the reference command for different attitude maneuvers.

The generated reference angle θ_{ref} is compared with the measured spacecraft angle to produce the tracking error. This error is supplied to a PID block configured as a PD controller with the parameters

$$K_p = 75.699, \qquad K_i = 0, \qquad K_d = 36.680$$

A derivative filter coefficient of

$$N = 100$$

was used to reduce derivative noise and improve practical implementation performance.

Simulation results confirmed that the step maneuver strategy successfully reduced overshoot during large-angle rotations while allowing the same controller gains to be used over the entire operating range. This avoided the need for gain scheduling and provided a simple, practical solution for achieving improved attitude pointing performance.